

EMS Readiness Optimization for Manhattan

Executive Presentation

Slide 1: Title

Optimizing Ambulance Staging in Manhattan

A Simulation-Based Approach to Reducing EMS Response Times

- EMS Optimization Research Team
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 - Prepared for FDNY Operations Leadership
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Slide 2: The Problem

Current EMS Response Performance is Unacceptable

The current uniform allocation policy (P0):

- Distributes ambulances **equally** across 48 firehouses
- **Ignores** where crashes actually happen
- Results in **3.17-minute average response time** at K=20
- **99.6%** of calls get response within 8 minutes (NFPA standard); **93.7%** within 6 minutes (NYC law)

Why this matters:

- Manhattan has ~3.48 MVC calls per hour (30,500+ per year)
- Midtown generates 3-5× more calls than Upper Manhattan
- Every minute of delay increases mortality risk for critical patients

“The difference between a 2-minute and 8-minute response can be the difference between life and death.”

Slide 3: Baseline Performance — P0 (Spatially-Stratified)

Even Geographic Spread Still Leaves Room for Improvement

Metric	P0 Value	Target
Mean Response Time	3.17 min	< 4 min
P95 Response Time	6.26 min	< 8 min
8-min Coverage	99.6%	> 95%
Utilization	7.6%	—

Context: The spatially-stratified baseline ensures geographic coverage across Manhattan, meeting the 8-minute target. However, it does not account for demand patterns, leaving room for response time improvements through demand-aware allocation.

- Midtown precincts (highest demand) could benefit from additional staging capacity
- A demand-proportional or optimized approach can further reduce average and tail response times

Slide 4: Our Solution — Optimized Allocation (P2)

Smart Staging: Put Ambulances Where They're Needed

Approach:

1. Analyzed **2.24 million** historical crash records
2. Built a demand model capturing hourly and geographic patterns
3. Used **mathematical optimization** to find the best allocation
4. Validated with **2,700+ simulation runs**

P2 Allocation Strategy:

- Concentrates units in **high-demand areas** (Midtown, Lower Manhattan)
- Maintains **minimum coverage** in low-demand areas
- Uses the **same 20 ambulances** — just staged differently
- Implementable as a **simple shift-change staging plan**

Slide 5: Key Results — Three-Policy Comparison

Progressive Improvement from P0 → P1 → P2

Metric	P0 (Baseline)	P1 (Demand-Proportional)	P2 (Optimized)
Mean Response Time	3.17 min	2.63 min	2.57 min
P95 Response Time	6.26 min	5.05 min	4.66 min
8-min Coverage	99.6%	99.6%	99.6%
Fleet Used	20 units	20 units	20 units

Interpretation:

- P1 (demand-proportional) delivers substantial gains over P0 by matching allocation to demand patterns
- P2 (optimized) provides additional tail-latency improvement, reducing P95 response time by 7.7% beyond P1
- All three policies use the same 20-unit fleet — differences are purely from staging decisions

Statistical confidence:

- All pairwise differences significant at $p < 0.001$
- Based on 30 independent simulation replications per policy

Slide 6: Performance Improvements Visualized

Response Time Distribution Shift

P0 Distribution: 3.17 min mean, compact distribution with P95 at 6.26 min

P2 Distribution: Tight cluster around 2.57 min, 99.6% under 8 min

Key Visual Insights (see results/figures/):

- `pub_fig1_policy_comparison.png` — Side-by-side policy comparison
- `pub_fig2_fleet_sensitivity.png` — How fleet size affects each policy
- `pub_fig5_performance_heatmap.png` — Performance across all scenarios

The bottom line: P2 doesn't just shift the average — it nearly eliminates the long waits that matter most.

Slide 7: Robustness Analysis

P2 Wins Under Every Condition We Tested

Fleet Size Sensitivity (K = 15 to 40):

- P2 achieves >99% coverage with just K=25 units

- P0 requires K=40 to reach 99% coverage
- **P2 with 15 units outperforms P0 with 40 units**

Demand Variation (0.5× to 2.0×):

- P2 mean RT ranges 2.44–2.85 min (stable)
- P0 mean RT ranges 3.00–3.50 min (modest degradation under stress)
- Policy rankings unchanged across all demand levels

Service Time Sensitivity (20–30 min mean):

- No impact on policy rankings
- Response time unaffected by service time variations
- Results hold under all operational assumptions tested

Slide 8: Implementation Roadmap

Three-Phase Deployment Plan

Phase 1 — Pilot (Months 1–3)

- Deploy P2 at **5 highest-impact firehouses** (Midtown)
- Monitor mean RT, 8-min coverage, and utilization daily
- Success criterion: $\geq 15\%$ improvement in pilot area response times

Phase 2 — Expansion (Months 3–6)

- Expand to **15–20 firehouses** based on pilot results
- Calibrate model with real dispatch data
- Begin road-network travel time integration

Phase 3 — Full Deployment (Months 6–12)

- Roll out to **all 48 Manhattan firehouses**
- Implement shift-specific allocations
- Establish continuous optimization cycle

Requirements: No new equipment, no new technology. Just repositioning existing ambulances.

Slide 9: Expected Benefits

Annual Impact Estimate (K=20 units)

Benefit	Value
Calls per year affected	~30,500 MVC calls
Mean RT reduction per call	0.60 minutes (3.17 → 2.57 min)
Total annual time saved	~18,300 minutes (~305 hours)
6-min coverage improvement	+4.2 pp (94.0% → 98.2%)
8-min coverage	99.7% — maintained at near-perfect levels

What this means in practice:

- Faster defibrillation and trauma response
 - Quicker stroke and cardiac care
 - Better coverage in currently underserved areas
 - Same performance with fewer units, or better performance with the current fleet
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Slide 10: Recommendations and Next Steps

Immediate Recommendations

1. **Approve P2 adoption** as the standard ambulance staging policy for Manhattan
2. **Launch pilot** at Midtown firehouses within 30 days
3. **Establish monitoring** dashboard with real-time KPI tracking
4. **Brief dispatch leadership** on new staging locations

Future Opportunities

- **Dynamic repositioning:** Adjust staging in real-time based on current unit positions
- **Multi-borough expansion:** Extend optimization to Brooklyn, Queens, Bronx, Staten Island
- **All-incident integration:** Include medical emergencies, fires, and hazmat calls
- **CAD integration:** Embed optimization recommendations in dispatch software

Questions?

Contact: EMS Optimization Research Team

Repository: github.com/cnsp/ems-optimization

Full Technical Report: [docs/technical_report.md](#)

This analysis is based on 2,700+ simulation experiments, 2.24 million historical crash records, and rigorous statistical validation. All results can be reproduced from the project repository.